

3

Level 3 Calculus

91578 Apply differentiation methods in solving problems

PRACTICE ASSESSMENT B (v6)

Achievement	Achievement with Merit	Achievement with Excellence
Apply differentiation methods in solving problems.	Apply differentiation methods, using relational thinking, in solving problems.	Apply differentiation methods, using extended abstract thinking, in solving problems.

You should attempt ALL the questions in this booklet.

Make sure that you have the Formulae and Tables Booklet L3–CALCF.

Show ALL working.

If you need more room for any answer, use the extra space provided at the back of this booklet.

Check that this booklet has pages 2–13 in the correct order and that none of these pages is blank.

QUESTION ONE

- (a) Differentiate $f(x) = (6 - x^3 + 2x)^4$.

You do not need to simplify your answer.

- (b) A curve is defined parametrically by the equations $x = 3 \tan(\theta)$ and $y = 2 \sin^2(\theta)$.

Find the rate of change of y with respect to x at $\theta = \frac{\pi}{4}$.

You must use calculus and show any derivatives that you need to find when solving this problem.

- (c) The line $y = ax + b + e$ is tangent to the curve $f(x) = \frac{e^x}{x} + 2x$ at $x = 1$, where a and b are constants. Find the value of a and b .

You must use calculus and show any derivatives that you need to find when solving this problem.

(d) The *Mean Value Theorem* states that provided a function $f(x)$ is continuous and differentiable for $a \leq x \leq b$, there must exist at least one value c between a and b such that $f'(c) = \frac{f(b)-f(a)}{b-a}$.

Given the function $f(x) = \frac{x^2}{x-1}$ defined between $x = 2$ and $x = 5$, find the coordinates of the point that is guaranteed to exist by this theorem.

You may assume that the function is continuous and differentiable between the x values given.

You must use calculus and show any derivatives that you need to find when solving this problem.

- (e) Consider the function $f(x) = e^{-x}(x^2 + kx + 1)$, where k is a real constant. The curve has exactly one stationary point.

A student claims that there might be a value of k such that the curve has no points of inflection.

Show that this claim is false. That is, show that the curve always has points of inflection.

You must use calculus and show any derivatives that you need to find when solving this problem.

QUESTION TWO

(a) Differentiate $f(x) = x - \frac{x}{\operatorname{cosec}(x)}$.

You do not need to simplify your answer.

(b) A marine biologist is analysing the carbon absorption rate of a newly engineered species of deep-sea kelp. The total amount of carbon absorbed, $C(t)$ in grams, after t days is modelled by the function $C(t) = 12(\ln(3t + 1))^2$.

Find the rate at which the kelp is absorbing carbon after 5 days.

You must use calculus and show any derivatives that you need to find when solving this problem.

(c) Consider the two functions: $f(x) = x \cos^2(\pi x)$ and

$g(x) = kx^2 - 3x$, where k is a constant.

At $x = 1$, the normal to the curve $f(x)$ is perpendicular to the normal to the curve $g(x)$, as shown in the diagram.

Find the equation of the tangent to the curve $g(x)$ at $x = 1$.

You must use calculus and show any derivatives that you need to find when solving this problem.

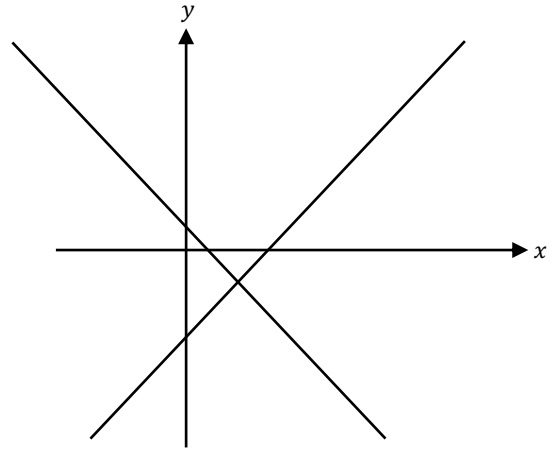
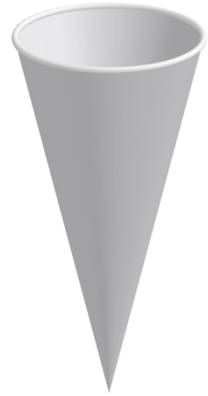


Diagram is NOT to scale

- (d) An inverted right circular conical tank is being filled with water at a constant rate of $18 \text{ cm}^3 \text{ s}^{-1}$. The tank has a maximum depth of 40 cm and a base radius of 10 cm at the top.

Find the rate at which the water level is rising when the depth of the water is 15 cm.

You must use calculus and show any derivatives that you need to find when solving this problem.



- forced to move backwards. Based on the speed and spin of the shot, the horizontal distance x , in metres, between the two players at the moment Ben attempts to intercept the ball can be modelled by the equation $x(\theta) = 3 \sin \theta \cos^2 \theta$. Ben can reach a maximum vertical height of 2.2 metres with his paddle.

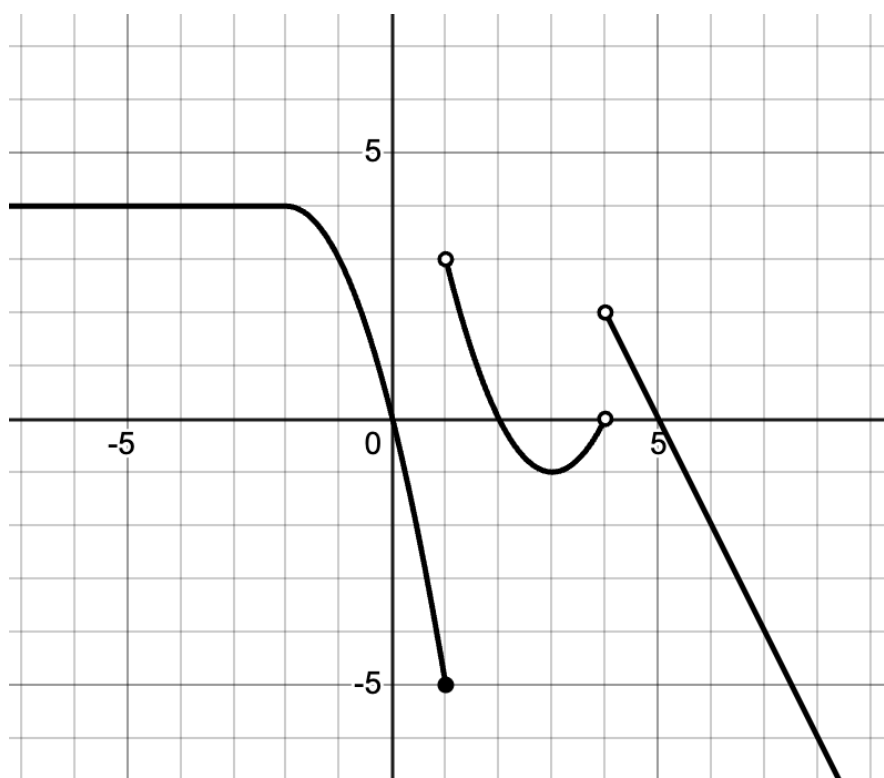


Show ALL working clearly. You may assume that the height you have found is a maximum.

You must use calculus and show any derivatives that you need to find when solving this problem.

QUESTION THREE

(a) The graph below shows the function $y = f(x)$.



For the function above:

(i) Find the value(s) of x where $f'(x) = 0$.

(ii) Find the value(s) of x where $f(x)$ is not differentiable.

(iii) Find the value of $\lim_{x \rightarrow 4} f(x)$. State clearly if the value does not exist.

(b) Consider the curve defined by the function $f(x) = \frac{4x}{e^{\frac{1}{8}x^2}}$.

Find the x -coordinate(s) of any stationary point(s) on the curve.

You do not need to determine the nature of the point(s).

You must use calculus and show any derivatives that you need to find when solving this problem.

(c) Consider the function $y = f(x) = \frac{x^2 + 2x + 1}{kx + 1}$, where k is a constant. The value of y always increases at the same rate as the value of x .

Find the value of k .

You must use calculus and show any derivatives that you need to find when solving this problem.

(d) Consider the curve defined by the pair of parametric equations:

$$x = (t^4 - 1)^3 \text{ and } y = (t^2 - 6t + 10)e^t.$$

A tangent to this curve is parallel to the x -axis. Find the equation of this tangent.

You must use calculus and show any derivatives that you need to find when solving this problem.

(e) Consider the functions $f(x) = 2x^2 - kx + 3$ and $g(x) = \frac{k}{x}$, where $x \neq 0$ and k is a real constant.

If $f(x) > g(x)$ for all values of x when $1 \leq x \leq 2$,

show that $k < \frac{5}{2}$.

You must use calculus and show any derivatives that you need to find when solving this problem.

